

Day 2 Lab Project: Build Something with Embeddings

Applied NLP · YDL 2026

Today you saw how to turn language into numbers: tokenization, the distributional hypothesis, Word2Vec embeddings, cosine similarity and analogies, tf-idf, and topic models. This project is your chance to **combine those ideas into something of your own** on data you actually care about. There is no single right answer; the goal is a small, complete thing that tells a story.

The one-sentence brief

Gather some text you find interesting, turn it into vectors using techniques from today, and produce one artifact (a plot, an interactive page, or a small app) that shows something you learned.

Main tasks

You do not need to turn in or work in the embeddings toolbox notebook (*lab_embeddings_toolbox_starter.ipynb*), but it has helpful runnable examples for every tool below.

- Use at least one technique from today on text you gathered yourself (not just the demo data).
- Show it works end to end: load text, build vectors, and inspect the result.
- Produce one artifact that tells a story: a labelled plot or an interactive HTML page.

Your toolbox (in the notebook)

- **Pretrained embeddings:** load GloVe/word2vec with gensim; nearest neighbors and analogies.
- **Train your own:** Word2Vec on any sequence of tokens (words, or the X2Vec trick: playlists, recipes, courses).
- **Visualize:** project vectors to 2D with PCA / t-SNE; write a standalone interactive scatter.
- **Classify:** tf-idf features vs averaged-embedding features with a simple classifier.
- **Get text:** helpers for Project Gutenberg, lyrics, and polite web scraping.

Project ideas (mix and match what we covered)

Each idea names the concepts it combines. **These are just suggestions, feel free to do your own thing, and own idea, so long as it makes use of one of the NLP technologies mentioned today.** Treat these as starting points, and bend them toward whatever you find fun.

Idea	What it is	Combines
X2Vec on something you love	Train embeddings on sequences from a domain you like: songs in playlists (Song2Vec), ingredients in recipes, courses in a schedule. Show the nearest neighbors make sense and plot the clusters.	Word2Vec + training + cosine + PCA/t-SNE viz

Idea	What it is	Combines
Semantic search engine	Embed a corpus (books, lyrics, a subreddit). Type a query, return the most similar passages by cosine similarity. Wrap it in the interactive HTML viz.	Embeddings + cosine + nearest neighbors
Author / genre / vibe classifier	Pick two or more classes (authors, genres, subreddits, sentiment). Compare tf-idf features against averaged-embedding features and report which wins and why.	tf-idf + embeddings + classification
Map of a corpus	Embed a collection you care about and build an interactive 2D map where distance means similarity. Color by category and explore the neighborhoods.	Embeddings as points + PCA/t-SNE + interactive viz
Analogy explorer / game	Probe analogies (king minus man plus woman). Find where they work and where they break; turn it into a tiny guessing game.	Word2Vec + vector arithmetic + analogies
Polysemy & bias probe	Investigate the limits of one vector per word: sense conflation ('bank'), or social bias baked into pretrained vectors. Write up what you find.	Limits of static embeddings + cosine
Topic-model a collection	Run topic modeling on books, news, or your own messages. Label the topics and, if you have timestamps, chart how topics rise and fall over time.	Topic models (LDA) + the Hall 2008 idea
Tokenizer investigation	Train BPE at several vocabulary sizes; measure OOV rate and sequence length. Compare your merges to a real LLM tokenizer in tiktokenizer.	Tokenization + BPE + types vs tokens
Custom vs pretrained embeddings	Train Word2Vec on a niche corpus (a fandom, a field, a language variety) and show it captures domain meanings that generic pretrained vectors miss.	Distributional hypothesis + training + cosine
Meaning through time	Train embeddings on old text (Project Gutenberg) vs modern text and show how a word's neighbors shift across eras.	Word2Vec + nearest neighbors + viz
Embedding recommender	Build a nearest-neighbor recommender (books, songs, courses): given one item, suggest similar ones from the embedding space.	X2Vec + cosine + nearest neighbors
Lyric / script mood map	Embed and cluster songs or movie lines; track sentiment across an album or film and visualize the arc.	Embeddings + clustering + viz

Where to find text data

A few reliable starting points, grouped by type. Always skim a source's terms of use and license before scraping or redistributing; prefer public-domain or openly-licensed text.

Dataset hubs (search here first)

Source	What / how to access	Notes
Hugging Face Datasets	180k+ datasets; one line: <code>load_dataset('ag_news')</code> , 'imdb', 'wikitext', etc. (huggingface.co/datasets)	Best first stop; filter by language/task/license
Kaggle Datasets	Thousands of community datasets, many text (kaggle.com/datasets)	Free account; quality varies
Google Dataset Search	Search engine across the web's datasets (datasetsearch.research.google.com)	Good for niche topics
niderhoff/nlp-datasets	Curated GitHub list of free/public-domain NLP corpora	Great for browsing by type

Books, encyclopedic, academic

Source	What / how to access	Notes
Project Gutenberg	70k+ public-domain books; search + download plain text via the Gutendex API (gutendex.com/books?search=...)	Public domain; great for literary text
Wikipedia	Wikimedia dumps (dumps.wikimedia.org), or HF 'wikitext' / 'wikipedia'	CC-licensed; clean prose
arXiv bulk data	Full text of research papers (arxiv.org/help/bulk_data)	Large; technical language
Google Books Ngrams	Word/phrase frequencies over centuries	Good for meaning-over-time

Social, forums, reviews

Source	What / how to access	Notes
Arctic Shift (Reddit)	Posts/comments as JSON via API + bulk dumps (arctic-shift.photon-reddit.com); the Pushshift successor	Use a download tool for one subreddit
Stack Exchange	Official data dumps (archive.org/details/stackexchange)	Q&A; text + tags
Amazon / Yelp / IMDb reviews	Amazon Reviews (McAuley, UCSD); Yelp Open Dataset; IMDb 50k (HF 'imdb')	Classic for sentiment; check license
AG News	News topic classification (HF 'ag_news')	Clean, labelled, four classes

Lyrics, film, dialogue

Source	What / how to access	Notes
lyrics.ovh	Free API, no key: <code>api.lyrics.ovh/v1/<artist>/<title></code>	Easiest for a quick lyric demo
Genius lyrics	Richer (lyricsgenius), or the bulk 'Genius Song Lyrics' dataset on Kaggle	API often rate-limited/blocked

Source	What / how to access	Notes
Cornell Movie-Dialogs	220k movie conversations (cs.cornell.edu) or HF 'cornell_movie_dialog'	Dialogue with metadata
IMSDb / OpenSubtitles	Movie scripts (Film Corpus 2.0); subtitles in many languages (opensubtitles.org)	Check terms before redistributing

Scraping your own text (when there is no API)

If a site has no API or dataset, you can scrape it with `requests` + `BeautifulSoup` (there is a template in the notebook). Do it responsibly:

- **Check robots.txt** (site.com/robots.txt) and read the site's terms of service before scraping.
- **Identify yourself** with a User-Agent string, and **rate-limit**: pause between requests (`time.sleep`) so you do not hammer the server.
- **Cache locally**: save what you fetch once, then work from disk; do not re-download repeatedly.
- **Respect copyright**: use text for your own analysis; do not redistribute scraped copyrighted content.

If a fetch fails

Many pages are fine; some block scripts. If a site blocks you, switch to a dataset hub or an API source above rather than fighting it; the point is the analysis, not the scrape.

Tips

Start small, get something working on a smaller piece of data, then scale up. Lean and focus on doing your hand implementations using the toolbox notebook's examples. However, for tasks like visualization, making unique looking aesthetics and so on, feel free to use Claude.

Sources for the data list: Hugging Face Datasets, Kaggle, Google Dataset Search, niderhoff/nlp-datasets, Gutendex / Project Gutenberg, Wikimedia, arXiv, Arctic Shift, Stack Exchange, Amazon Reviews (McAuley/UCSD), Yelp Open Dataset, IMDb, AG News, lyrics.ovh, Genius, Cornell Movie-Dialogs, IMSDb, OpenSubtitles.